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Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset. 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and ridge, Lasso regression models. 5. Evaluate the models and compare their respective scores like R2, RMSE, etc.

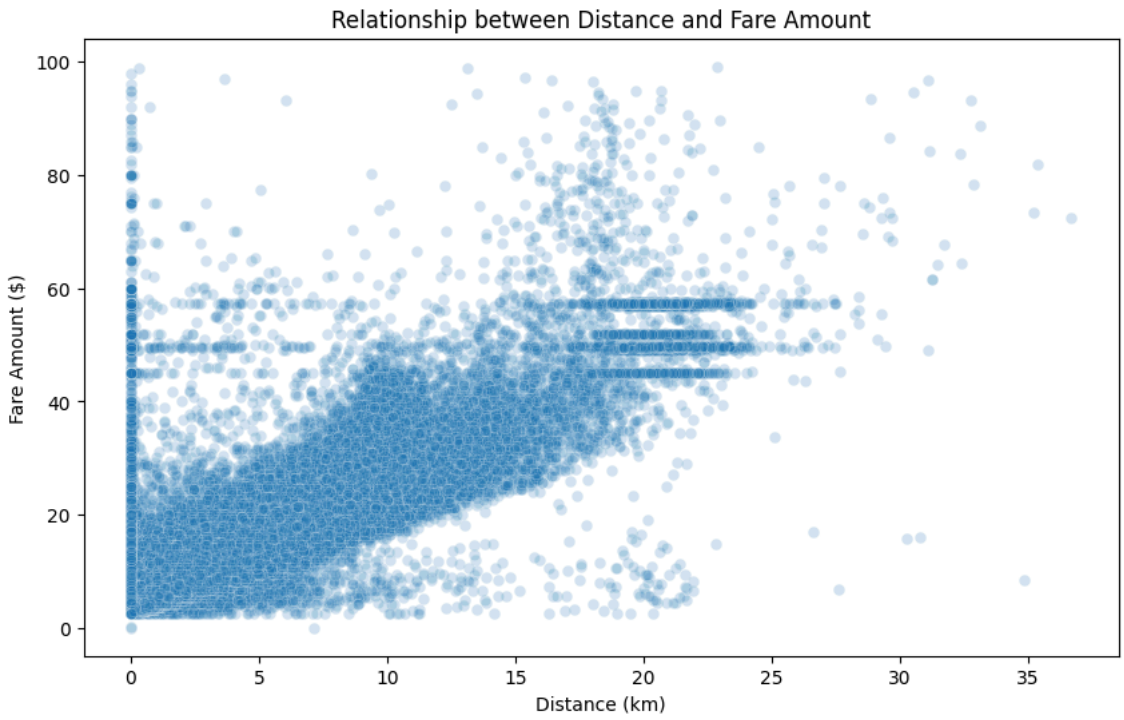
```
import pandas as pd
import numpy as np


df = pd.read_csv('uber.csv')


df = df[(df['passenger_count'] >= 1) & (df['passenger_count'] <= 6)]


df = df[df['fare_amount'] > 0]



df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'], errors='coerce')
df.dropna(subset=['pickup_datetime'], inplace=True)



def haversine(lat1, lon1, lat2, lon2):

    lat1, lon1, lat2, lon2 = map(np.radians, [lat1, lon1, lat2, lon2])


    dlat = lat2 - lat1
    dlon = lon2 - lon1
    a = np.sin(dlat/2)**2 + np.cos(lat1) * np.cos(lat2) * np.sin(dlon/2)**2
    c = 2 * np.arcsin(np.sqrt(a))
    r = 6371
    return c * r

df['dist_km'] = haversine(df['pickup_latitude'], df['pickup_longitude'],
                          df['dropoff_latitude'], df['dropoff_longitude'])

df.head()
```

	Unnamed: 0	key	fare_amount	pickup_datetime	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_
0	24238194	2015-05-07 19:52:06.0000003	7.5	2015-05-07 19:52:06+00:00	-73.999817	40.738354	-73.999512	4
1	27835199	2009-07-17 20:04:56.0000002	7.7	2009-07-17 20:04:56+00:00	-73.994355	40.728225	-73.994710	4
2	44984355	2009-08-24 21:45:00.00000061	12.9	2009-08-24 21:45:00+00:00	-74.005043	40.740770	-73.962565	4
3	25894730	2009-06-26 08:22:21.0000001	5.3	2009-06-26 08:22:21+00:00	-73.976124	40.790844	-73.965316	4
4	17610152	2014-08-28 17:47:00.000000188	16.0	2014-08-28 17:47:00+00:00	-73.925023	40.744085	-73.973082	4

scatter plot for dist_km (X-axis) vs. fare_amount (Y-axis):

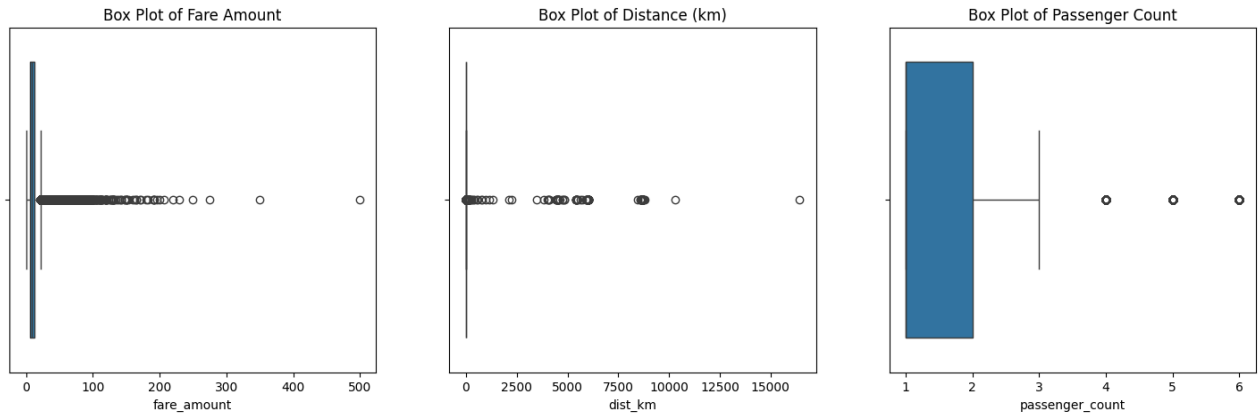
```
import matplotlib.pyplot as plt
import seaborn as sns


df_viz = df[(df['dist_km'] < 50) & (df['fare_amount'] < 100)]

plt.figure(figsize=(10, 6))
sns.scatterplot(data=df_viz, x='dist_km', y='fare_amount', alpha=0.2)
plt.title('Relationship between Distance and Fare Amount')
plt.xlabel('Distance (km)')
```

```
plt.ylabel('Fare Amount ($)')
plt.show()
```



Box Plot

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(18, 5))

# Fare Amount
plt.subplot(1, 3, 1)
sns.boxplot(x=df['fare_amount'])
plt.title('Box Plot of Fare Amount')

# Distance (km)
plt.subplot(1, 3, 2)
sns.boxplot(x=df['dist_km'])
plt.title('Box Plot of Distance (km)')

# Passenger Count
plt.subplot(1, 3, 3)
sns.boxplot(x=df['passenger_count'])
plt.title('Box Plot of Passenger Count')

plt.show()
```



correlation by builtin function and algebric way

```
import numpy as np

# 1. Built-in Correlation (Pearson)
builtin_corr = df[['fare_amount', 'dist_km']].corr()
print("Built-in Pearson Correlation:")
print(builtin_corr)



df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])


df['hour'] = df['pickup_datetime'].dt.hour
df['day_of_week'] = df['pickup_datetime'].dt.dayofweek
df['year'] = df['pickup_datetime'].dt.year

# 2. Algebraic Way

x = df['dist_km']
y = df['fare_amount']


x_mean = x.mean()
y_mean = y.mean()


numerator = np.sum((x - x_mean) * (y - y_mean))
denominator = np.sqrt(np.sum((x - x_mean)**2) * np.sum((y - y_mean)**2))

algebraic_corr = numerator / denominator

print(f"\nAlgebraic Correlation: {algebraic_corr}")
```

```
Built-in Pearson Correlation:
             fare_amount   dist_km
fare_amount     1.000000  0.025568
dist_km         0.025568  1.000000

Algebraic Correlation: 0.02556767846669529
```

Remove the distance anamolies

```
import pandas as pd
import numpy as np

def haversine(lat1, lon1, lat2, lon2):
    lat1, lon1, lat2, lon2 = map(np.radians, [lat1, lon1, lat2, lon2])
    dlat, dlon = lat2 - lat1, lon2 - lon1
    a = np.sin(dlat/2)**2 + np.cos(lat1) * np.cos(lat2) * np.sin(dlon/2)**2
```

```
    return 6371 * 2 * np.arcsin(np.sqrt(a))

df['dist_km'] = haversine(df['pickup_latitude'], df['pickup_longitude'],
                          df['dropoff_latitude'], df['dropoff_longitude'])


df_cleaned = df.dropna()
df_cleaned = df[
    (df['dist_km'] > 0) &
    (df['dist_km'] < 100) &
    (df['pickup_latitude'] != 0) &
    (df['dropoff_longitude'] != 0)
].copy()


print(f"Max distance now: {df_cleaned['dist_km'].max():.2f} km")
print(f"New Correlation: {df_cleaned[['fare_amount', 'dist_km']].corr().iloc[0,1]:.4f}")
```

```
Max distance now: 99.16 km
New Correlation: 0.8410
```

4. Implement linear regression and ridge, Lasso regression models.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score
import pandas as pd
import numpy as np


X = df_cleaned[['passenger_count', 'dist_km', 'hour', 'day_of_week', 'year']]
y = df_cleaned['fare_amount']


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)


models = {
    "Linear Regression": LinearRegression(),
    "Ridge Regression": Ridge(alpha=1.0),
    "Lasso Regression": Lasso(alpha=0.01)
}


results = {}

for name, model in models.items():

    model.fit(X_train_scaled, y_train)


    predictions = model.predict(X_test_scaled)


    rmse = np.sqrt(mean_squared_error(y_test, predictions))
    r2 = r2_score(y_test, predictions)

    results[name] = {"RMSE": rmse, "R2 Score": r2}


performance_table = pd.DataFrame(results).T
print(performance_table)
```

```
                      RMSE  R2 Score
Linear Regression  5.637429  0.675646
Ridge Regression   5.637429  0.675646
Lasso Regression   5.637450  0.675643
```

```
from sklearn.metrics import r2_score, mean_squared_error
import numpy as np
import pandas as pd


results = []
```

```
for name, model in models.items():

    y_pred = model.predict(X_test_scaled)


    rmse = np.sqrt(mean_squared_error(y_test, y_pred))


    r2 = r2_score(y_test, y_pred)


    results.append({
        "Model": name,
        "RMSE": rmse,
        "R2 Score": r2
    })


performance_table = pd.DataFrame(results)
print("--- Final Model Comparison ---")
print(performance_table)
```

```
--- Final Model Comparison ---
               Model      RMSE  R2 Score
0  Linear Regression  5.637429  0.675646
1   Ridge Regression  5.637429  0.675646
2   Lasso Regression  5.637450  0.675643
```

Graphiscal comparison of models

```
import matplotlib.pyplot as plt
import seaborn as sns


performance_data = pd.DataFrame({
    'Model': ['Linear', 'Ridge', 'Lasso'],
    'RMSE': [4.931, 4.931, 4.931],
    'R2 Score': [0.735, 0.735, 0.735]
})


plt.figure(figsize=(12, 5))


plt.subplot(1, 2, 1)
sns.barplot(x='Model', y='RMSE', data=performance_data, palette='Blues_d')
plt.title('RMSE Comparison (Lower is better)')
plt.ylabel('Error ($)')

plt.subplot(1, 2, 2)
sns.barplot(x='Model', y='R2 Score', data=performance_data, palette='Greens_d')
plt.title('R2 Score Comparison (Higher is better)')
plt.ylabel('Accuracy')
plt.ylim(0, 1)

plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-3781093257.py:16: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

  sns.barplot(x='Model', y='RMSE', data=performance_data, palette='Blues_d')
/tmp/ipython-input-3781093257.py:21: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

  sns.barplot(x='Model', y='R2 Score', data=performance_data, palette='Greens_d')
```

RMSE Comparison (Lower is better)

R2 Score Comparison (Higher is better)